



441/2

AGRICULTURE PAPER 2

MARKING SCHEME

1. Two meal goals in Kenya

- (i). Galla
- (ii). Boer.

2. Function of

- (i) Testes – produce sperms
 - produce male hormone – testosterone
- (ii). Urethra – part of urinary system / conveys urine
 - Expulsion of sperms

3. Factors affecting digestibility

- Chemical composition of feed
- Form in which offered
- The ratio of energy to protein the higher the ratio the lower the digestibility
- Amount of feed already present in the digestive system.

4. Factors considered when selecting egg for sale.

- Cleanliness
- Egg size
- Egg colour
- Candling quality
- Shape of egg
- Smooth without cracks

5. Two uses of a gear box in a tractor .

- Provision of different forward speeds
- Enables reversing
- Allows tractor to stop without switching off the engine

6. Differences between out crossing and cross – breeding .**(2mrks)**

- Out crossing – mating of unrelated animals within same breed
- Cross – breeding –mating of two animals from different breeds.

7. Reason for weighing livestock at weaning**(1½ mrk)**

- Determine the amount of food to give
- Determine dosage of medicine
- Determine growth rate
- For selection / culling
- For record keeping

8. Four function of vitamins

- Blood clotting
- Help in muscular activity
- Act as organic solvents in metabolic reactions
- Development of strong bones
- Promote growth
- Protection against diseases (4 x ½ = 2mks)



9. Advantage of wool as construction material

(2mrks)

- Easily available / locally available
- Cheaper than most materials e.g. concrete
- Light to transport
- Requires less skills in use and preparation

10. Safety precautions when using electric power

(2mrks)

- Proper installation by qualified personnel
- Do not overload
- Ensure proper current rating according to need
- Avoid contact with naked wires
- Avoid contact with water
- Have appropriate / proper fire extinguishers (not use water / from for extinguishing electrical fires).

11. Tractor drawn implements attached to the drawn bar

- Trailors
- harrows

12. Uses of footbath in plunge dip

(2mrks)

- remove mud from hooves
- control foot rot / harden hooves where formalin is added.

13. Channels of marketing livestock

(2mrks)

- Local slaughtlin houses / Butcheries
- Kenya meat commission
- Livestock marketing Division

14. Disadvantages of using hurricane / lanterns to provide heat in a brooder

(2mrks)

- Provide inadequate and irregular heat.
- Risky as paraffin may spill and cause fire

15. Four ways of stimulating milk let down in a dairy cow.

(2mrks)

- Washing with warm water
- Familiar noises
- Sucking from calf
- Feeding concentrates
- Presences of milk man
- Presences of making utensils
- Milking at regular times

16. Function of additions in silage making

(2mrks)

- Improve carbohydrate level for proper fermentation
- Improve quality / palatability of silage

17. Uses of

- Chipping hammer
 - Removing rough stone surfaces / shaping rough stone surface.
- Tin snip – cutting sheet metal
- Router – removing wood to form a groove / make groove smooth.

18. (a). identification of

- A- Lice
- B- Flea
- C- Liver fluke / fasciola SPP.
- D- Roundworm / Ascaris SPP

(b).Differences

- A. and B are External / Ectoparasites
- C and D are internal parasites / Endoparasites

(c).Effective control mellird of C



- Eradicate water snail / round snail / lymnac SPP
- Deworming / use of Antihelruintics
- Draining marshy areas,

19. (a). Farm implement – ox – plough

(1/2mrk)

(b). A- Mould board

(2 $\frac{1}{2}$ mrks)

B – Share

C – Main beam

(D) – Land – wheel

(E) – Land – side

(c). Function of .

(1mrk)

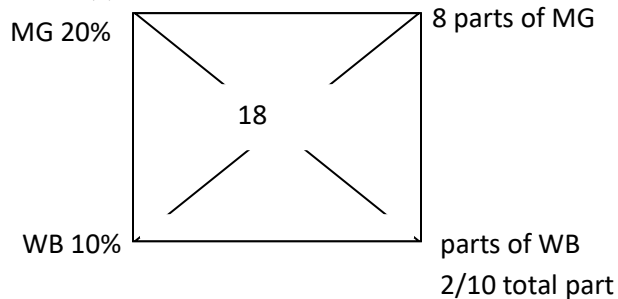
C – Attachment of all parts.

-Adds weight for deeper ploughing`

E – Stabilizes plough against thrust by furrow slices

(1mrk)

20. (a).Calculation



Calculating for

$$MG = \frac{8}{10} \times 100 = 80\text{kg}$$

$$WB = \frac{2}{10} \times 100 = 20\text{kg}$$

$$\frac{2}{10} \text{ parts of WB}$$

Total parts

(b). Other methods

- Trial and error

21. Stages of four stroke cycle engine

- G –
- H –
- I –
- J –

(b) Disadvantages of a four stroke engines

(2mrks)

- Are expensive to buy and maintain
- Their use is limited in areas
- They require skilled personal and support services

(c) .Function of part K

- Produce spark for ignition

SECTION C

22. (a). Importance of keeping livestock healthy.

(8mrks)

- Healthy animals grow well and fast reaching maturity quickly
- Healthy animals have longer economic and productive life
- Healthy animals give maximum production
- Healthy animals give good quality products and hence high market prices
- Healthy animals will not spread diseases to other animals and human beings



- Healthy animals are economical and easy to keep as the farmer spends less money on disease treatment hence reduced cost of production
- Have high fertility rate and produce more and faster
- Command high market prices / value eg beef animals

(b). Daily maintenance and servicing of a tractor.

(10mrks)

- Check engine oil using dip stick and adjust accordingly
- Check fuel level
- Check water level in radiator
- Check level of electrolyte in the battery
- Check for loose nuts and bolts and tighten
- Grease moving parts,
- Check tyre pressure
- Check and remove sediments in sediment owl
- Check fan belt tension and adjust accordingly
- Check the breaks and maintain break fluid level on recommended

(c). Maintenance practices in a fish pond

(2mrks)

- Fertilize fish pond – to increase planktons (food)for fish
- Remove silt – to maintain high oxygen concentrations in pond for survival of fish
- Liming – remove acidity from organic wastes
- Repair cracks in pond - to maintain water level

23. Structural requirements in construction of a calf pen.

(10mrks)

- Concrete floor – for easy clearing
- Spacious – to allow exercise and placement of equipment
- Singly – crawl spread of parasites
-preventing licking one and their to control formation of hair balls
- Proper lighting – Allow enough light
- For vitamin D synthesis
- Proper drainage – prevent dampness which predispose to infections.
- Drought free - to prevent cold winds which predispose to pneumonia
- Leak proof - Avoid damp conditions / wetness which predispose to navel ill; pneumonia
- Warm + dry – to avoid infections
- Well ventilated – Allow proper air circulation in structure
- Lockable / secure – provide security against predators / thieves

(b). Mastitis disease under

(i). Two causal organisms

- Streptococcus agalatiac (1mrk)
- Staphylococcus aureus (1mrk)

(ii). Predisposing factors

(8mrks)

- Sex
- Age – older lactating animals more prove
- Stage of lactation period – majorly during early /
- Under attachment – milk is good medicine for bacterial growth
- Mechanical injury - leads to injection
- Poor sanitation / hygiene – leads to infection
- Poor milking technique – Destroy sprinkler muscle of teal
- Hereditary factors

24. (a). Factors to consider while sitting farm structures

(10mrks)

- Location of homestead – centrally placed
- Accessibility
- Drainage
- Direction of preventing wind



- Relationship between structures
- Topography
- Proximity of amenities e.g. water / electricity
- Government policy

(b). Training a calf to drink milk from a bucket

(5mrks)

- Put clean milk in bucket
- Milk should be at body temperature / warm
- Place index finger into the calf's mouth, the calf starts sucking
- Ensure nails are short and clean
- Lower fingers slowly until it is submerged in the milk as the calf sucks. This allows the calf to drink milk
- Slowly withdraw the finger which calf is sucking
- Repeat the above procedure until calf can take milk on its own from bucket.

(c). Why prefer dric plough over mould board plough.

- Can easily be used on field with obstacles / rolls over obstacles / does not break.
- Requires less power to pull when operating
- Cuts at varying depths while poughing
- Wears out move slowly (drics) as compared to the share in mould board.
- May be the only one available